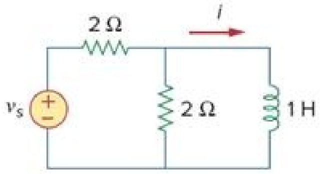
Chapter 18, Problem 45P

Problem

Use the Fourier transform to find *i*(*t*) in the circuit of Fig. 18.47 if *vs*(*t*) = 10*e*−2*tu*(*t*).

Figure 18.47



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

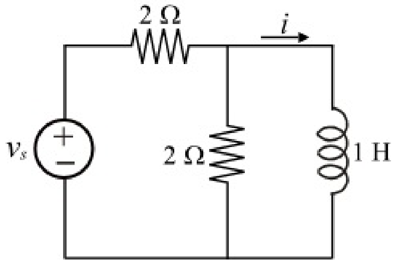


Figure 1

Step 2 of 5

Consider the input source voltage, $v_s(t) = 10e^{-2t}u(t)$.

Redraw the circuit in Figure 1 after converting the voltage source into a current source.

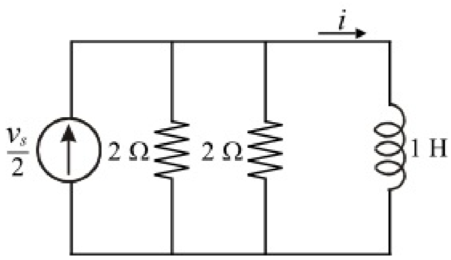


Figure 2

Step 3 of 5

Redraw the circuit in Figure 2 by replacing the two parallel resistances with their equivalent $1\,\Omega$ resistance.

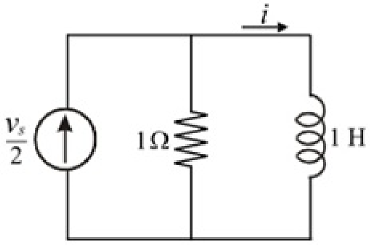


Figure 3

Step 4 of 5

Convert the time domain function v_s into the frequency domain using the Fourier transform pair.

$$\begin{aligned} V_s &= \mathcal{F}(v_s) \\ &= \mathcal{F}\left[10e^{-2t}u(t)\right] \\ &= 10\mathcal{F}\left[e^{-2t}u(t)\right] \qquad \left(\because e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a+j\omega}\right) \\ &= \frac{10}{j\omega+2} \end{aligned}$$

Draw the Figure 3 in Frequency domain.

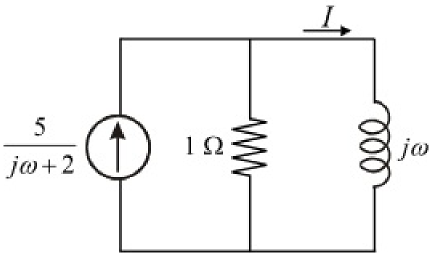


Figure 4

Step 5 of 5

Calculate the current I in Figure 4 using the current division formula.

$$\begin{aligned}
I &= \left(\frac{5}{j\omega+2}\right)\left(\frac{1}{1+j\omega}\right)\\
&= \left(\frac{5}{(j\omega+2)(1+j\omega)}\right)\\
&= \frac{A}{j\omega+2}+\frac{B}{1+j\omega}
\end{aligned}$$

Here *A* and *B* are constants.

Calculate the constant value *A*, using the residue method.

$$\begin{aligned}
A &= (j\omega+2)I\big|_{j\omega=-2}\\
&= (j\omega+2)\left(\frac{5}{(j\omega+2)(1+j\omega)}\right)\Bigg|_{j\omega=-2}\\
&= \left(\frac{5}{(-2+1)}\right)\\
&= -5
\end{aligned}$$

Calculate the constant value *B* using the residue method.

$$\begin{aligned}
B &= (j\omega+1)I\big|_{j\omega=-1}\\
&= (j\omega+1)\left(\frac{5}{(j\omega+2)(j\omega+1)}\right)\Bigg|_{j\omega=-1}\\
&= \left(\frac{5}{(2-1)}\right)\\
&= 5
\end{aligned}$$

Substitute obtained *A* and *B* values in equation I.

$$\begin{aligned}
I &= \frac{(-5)}{j\omega+2}+\frac{5}{1+j\omega}\\
&= \frac{5}{1+j\omega}-\frac{5}{2+j\omega}
\end{aligned}$$

Apply the inverse Fourier transform to find $i(t)$

$$\begin{aligned}
i(t) &= \mathcal{F}^{-1}(I)\\
&= \mathcal{F}^{-1}\left[\frac{5}{(j\omega+1)}-\frac{5}{(j\omega+2)}\right] \qquad \left(\because\ e^{-at}u(t)\xleftrightarrow{\mathcal{F}}\frac{1}{a+j\omega}\right)\\
&= 5e^{-t}u(t)-5e^{-2t}u(t)\\
&= 5\left(e^{-t}-e^{-2t}\right)u(t)\ \text{A}
\end{aligned}$$

Thus, the current $i(t)$ in the circuit of Figure 1 is $\boxed{5\left(e^{-t}-e^{-2t}\right)u(t)\ \text{A}}$.